

COMP 2310 - Lab 4

1- Let $A = \{-5, -4, -3, -2, -1, 0, 1, 2, 3, 4, 5\}$. (15 points)

$$R_1 = \{(a, b) : a \leq b\}$$

$$R_2 = \{(a, b) : a^3 = b^3\}$$

$$R_3 = \{(a, b) : a \times b \leq 0\}$$

a) For each of the above relations on A , determine if it is reflexive, symmetric, antisymmetric, or transitive.

(Justify / prove your answer for each item)

b) Which relation is partial order and which one is an equivalence relation? (why)

c) If a relation is partial order then draw its Hasse diagram.

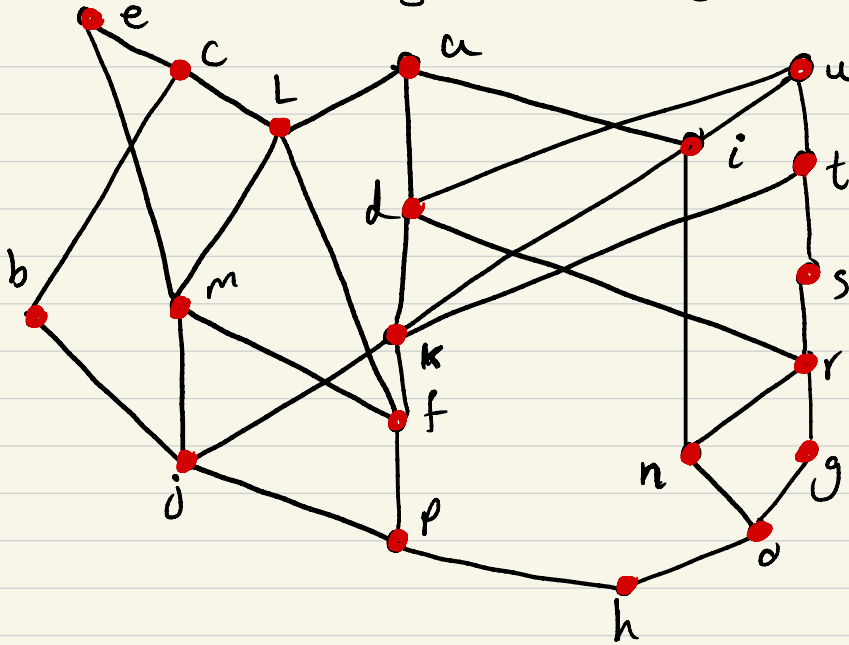
2- Let R be a relation on $\mathbb{N}$ such that $x R y : 3 \mid x - y$.

Prove that R is an equivalence relation.

Identify the equivalence classes of R .

(8 points)

3- Find the following items in the given Hasse diagram. (10 points)



a) $ub(\{m, k, t\})$

b) $lb(\{c, d, t\})$

c) $glb(\{u, k, m\})$

d) $lub(\{b, f\})$

e) $glb(\{p, k\})$

f) $glb(lub(\{n, k, j\}))$

4 - Using set identities and the fact that $x - y = x \cap \bar{y}$ prove that $(A - B) - (B - C) = A - B$. (8 points)

5 - Using set-builder notation prove that

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C) \quad (5 \text{ points})$$